

Convex Mixed-Integer Programming for Causal Additive Models with Optimization and Statistical Guarantees

Xiaozhu Zhang*, Nir Keret[†], Ali Shojaie*[†], Armeen Taeb*

Department of Statistics, University of Washington*; Department of Biostatistics, University of Washington[†]



Goal

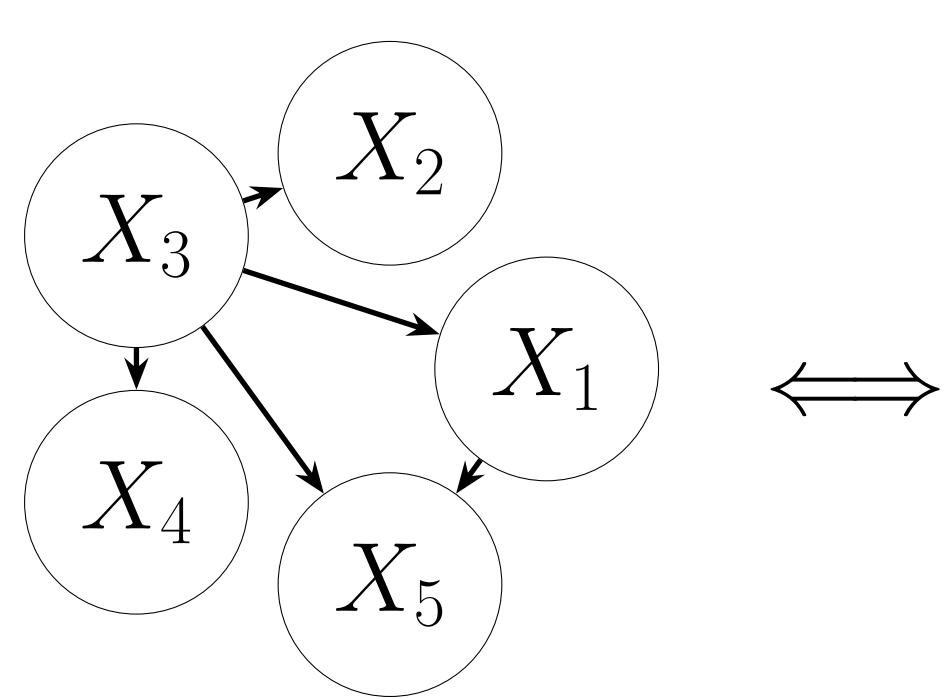
Causal discovery (DAG) for **non-linear** models:

- with **efficiency**
- with **optimality** guarantees
- with **statistical** guarantees
- without equal-variance assumption

Weaknesses of existing methods:

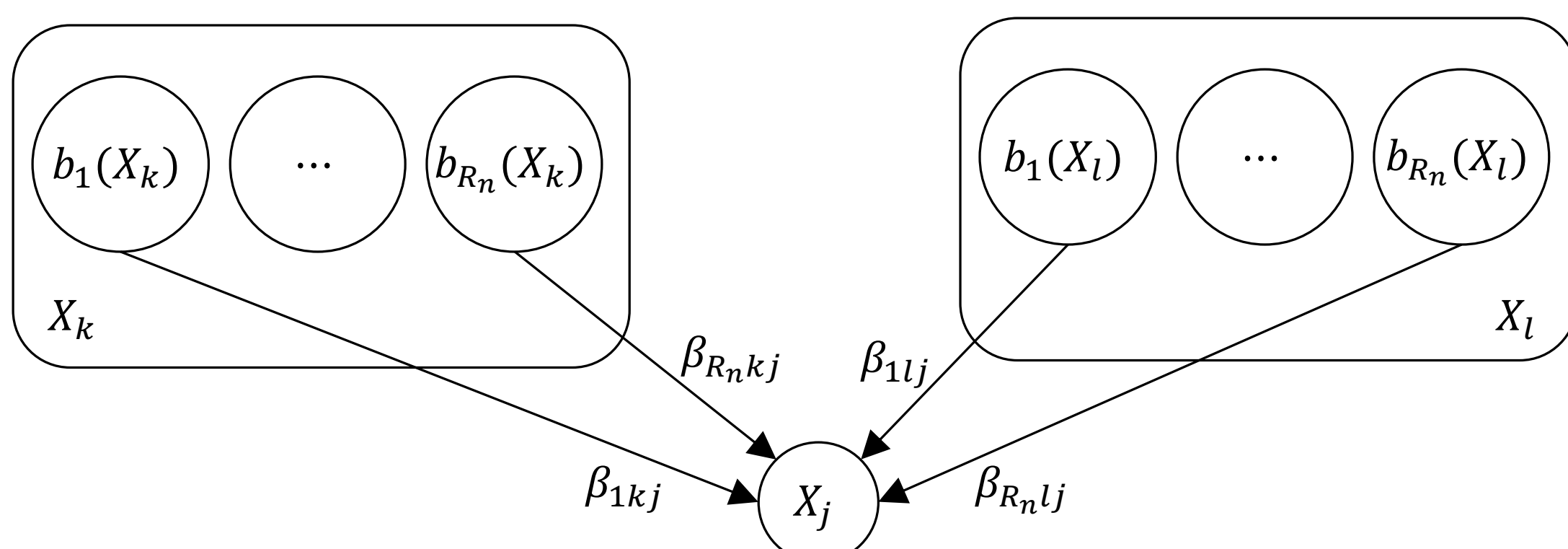
- CAM^[1], NoTears^[2]: no optimality guarantees
- NPAR^[3]: equal-variance assumption

Model: additive SEM with Gaussian errors



$$\begin{aligned} \text{For } j = 1, \dots, p, \\ X_j &= \sum_{k \in \text{pa}(j)} f_{kj}^*(X_k) + \epsilon_j, \\ \epsilon_j &\sim \mathcal{N}(0, \sigma_j^{*2}), \quad \sigma_j^* > 0, \\ \mathbb{E}[f_{kj}^*(X_k)] &= 0, \quad \forall j, k \end{aligned}$$

- **Faithfulness**: $f_{kj}^* \not\equiv 0$ if and only if $k \in \text{pa}(j)$
- **Identifiability of DAG**: when f_{kj}^* are non-linear and in C^3
- **Using basis functions**: $f_{kj}(x_k) = \sum_{r=1}^{R_n} \beta_{rkj} b_r(X_k)$



$$Z_j = \begin{bmatrix} b_1(X_1) & \dots & b_1(X_p) & \dots & b_{R_n}(X_1) & \dots & b_{R_n}(X_p) \end{bmatrix} \in \mathbb{R}^{n \times p R_n}$$

$$\beta_j = \begin{bmatrix} \beta_{11j} & \dots & \beta_{1pj} & \dots & \beta_{R_n 1j} & \dots & \beta_{R_n pj} \end{bmatrix} \in \mathbb{R}^{p R_n}$$

- Group sparsity: If $k \not\rightarrow j$, then $\beta_{rkj} = 0, \forall r$
- Negative log-likelihood:

$$\ell_n(\beta, \sigma) = \sum_{j=1}^p \log \sigma_j^2 + \sum_{j=1}^p \frac{\|X_j - Z_j \beta_j\|_n^2}{\sigma_j^2}$$

A mixed-integer programming (MIP) formulation

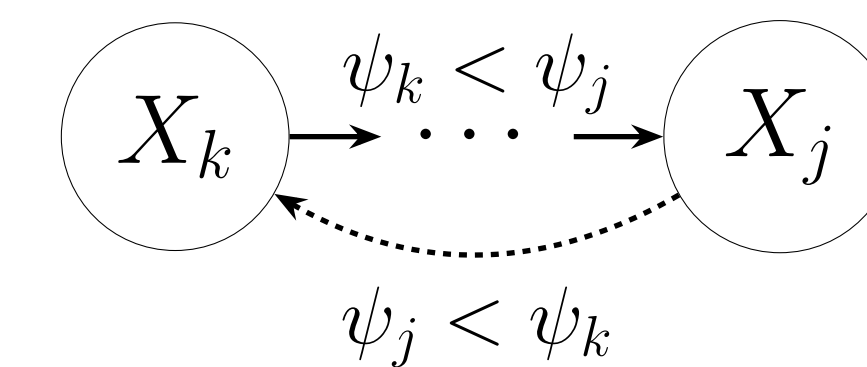
- **Edge selection constraint**:

$$\begin{aligned} \text{Edge presence } g_{kj} \in \{0, 1\}: \quad g_{kj} = 1 &\iff k \rightarrow j \\ g_{kj} = 0 &\implies \beta_{rkj} = 0, \quad \forall r \end{aligned}$$

- **Acyclicity constraint**:

$$\text{Layer value } \psi_j \in [1, p]: \quad k \rightarrow j \implies \psi_k < \psi_j$$

$$1 - p + p g_{kj} \leq \psi_j - \psi_k$$



- **Regularization**: $\lambda_n \sum_{k,j} g_{kj}$ (DAG sparsity); ridge penalty on β

The MIP formulation

g : binary β : continuous

$$\begin{aligned} \min_{\beta, g, \psi} \quad & \sum_{j=1}^p \log \left[\|X_j - Z_j \beta_j\|_n^2 + \gamma \|\beta_j\|_2^2 \right] + \lambda_n \sum_{k,j} g_{kj}, \\ \text{s.t.} \quad & (1 - g_{kj}) \beta_{rkj} = 0, \quad \forall r, k, j \quad (\text{edge selection}) \\ & 1 - p + p g_{kj} \leq \psi_j - \psi_k, \quad \forall k, j \quad (\text{acyclicity}) \end{aligned}$$

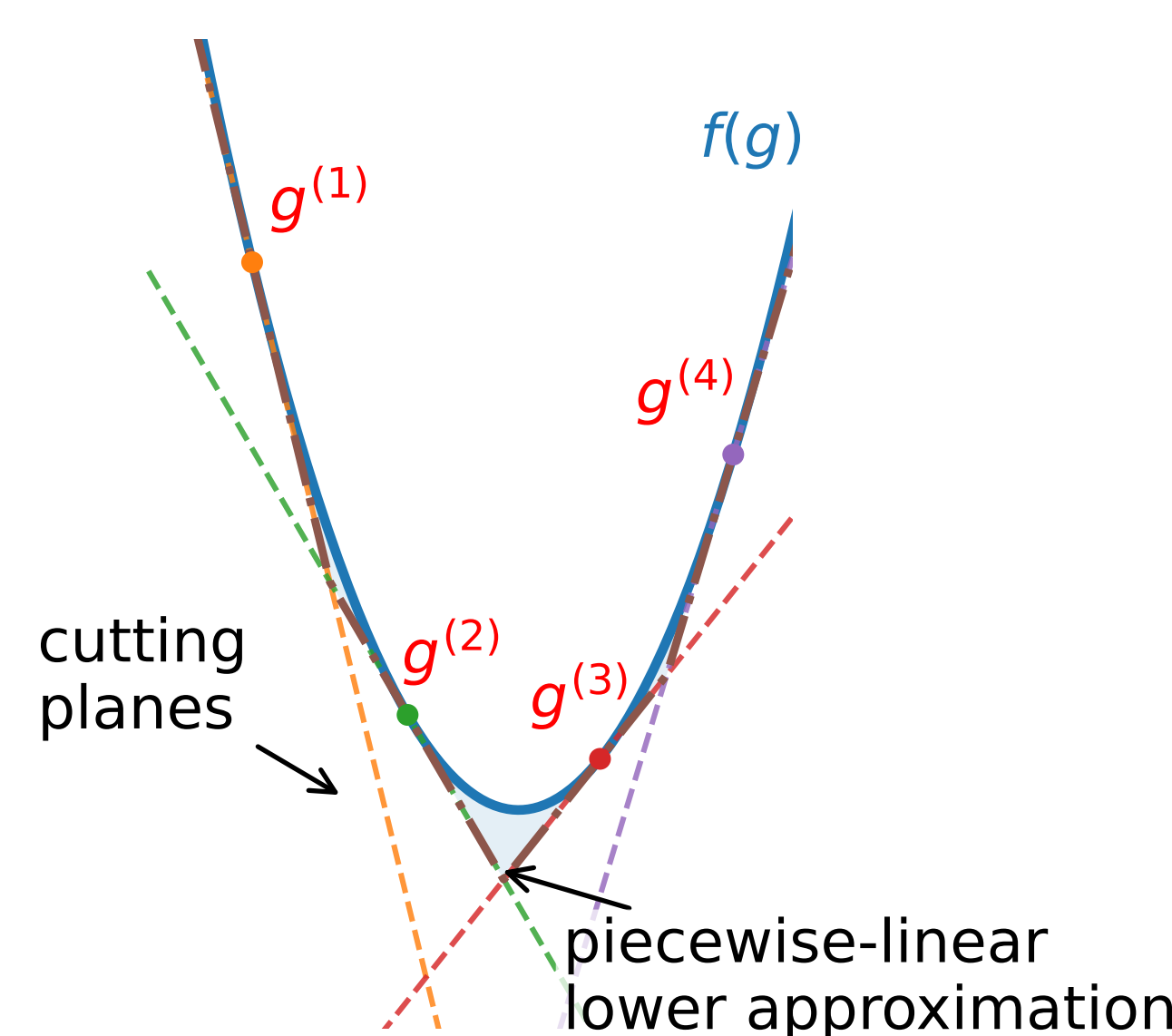
A novel outer approximation (OA) procedure

- Literature: OA for quadratic MIP
- We derive a novel OA for our non-quadratic MIP w/ the **log term**
 - We prove

$$f(g) = \min_{\beta} F(\beta, g) \quad \text{s.t. edge selection constraint}$$

can be solved in **closed-form** and $f(g)$ is **convex**

- We solve $\min_g f(g)$ s.t. acyclicity constraint via iterative cuts



u : feasible $f(g)$

l : lower approximations

Early stop when $|u - l| \leq \tau^{\text{early}}$

A linear MIP in g (scalable)

$$\begin{aligned} \min_{\theta, g, \psi} \quad & \theta + \lambda_n \sum_{k,j} g_{kj} \\ \text{s.t.} \quad & 1 - p + p g_{kj} \leq \psi_j - \psi_k, \quad \forall k, j, \\ & \theta \geq f(g^{(i)}) + \nabla f(g^{(i)})^\top (g - g^{(i)}), \\ & \forall \text{ cuts } i \end{aligned}$$

Theoretical guarantees

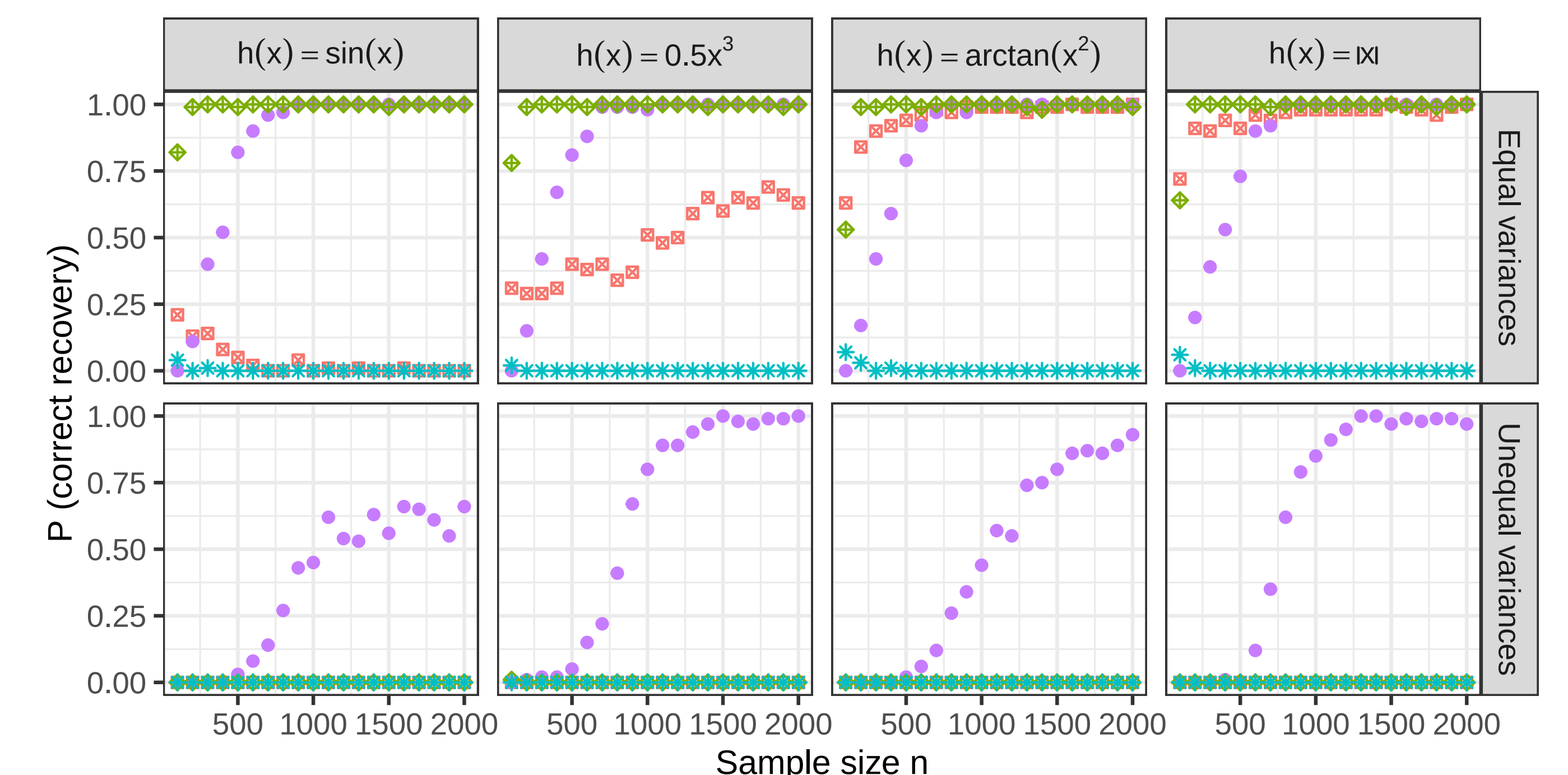
Accommodate **non-linearity** and **regularization** simultaneously

Finite-sample recovery

Under certain regularity conditions, with a proper choice of λ_n , when n is sufficiently large, it holds, with probability at least $1 - 6/p$:

- 1) The optimal **estimated graph** equals the **true graph**;
- 2) The result holds even with **early stopped estimated graph** when τ^{early} is sufficiently small.

Experiments



Method: CAM-IncEdge, NPVAR, MIP-linear, MIP-nonlinear (our approach)

Fig.: Graph recovery performance of baseline methods and our method on a five-variable example. The function $h(x)$ determines the type of non-linearity.

Dataset. p, s^*	MIP OA		CAM		NPVAR	
	Time	SHD	Time	SHD	Time	SHD
dsep.6.6	0.10	0.3	1.62	2.3	0.14	8.5
asia.8.8	0.22	0.5	2.38	0.0	0.30	12.1
bowling.9.11	0.32	1.1	2.93	1.6	0.42	14
inssmall.16.25	2.18	5.1	7.51	12.5	2.07	36.2
insurance.27.52	56.17	8.3	15.58	16.7	15.3	82

Table: Performance comparison when the SEM is unequal-variance.

References

- [1] Bühlmann, P., Peters, J. and Ernest, J. (2014), 'CAM: Causal additive models, high-dimensional order search and penalized regression', The Annals of Statistics 42(6), 2526 – 2556.
- [2] Seng, J., Zecevic, M., Dhani, D. S. and Kersting, K. (2022), 'Tearing apart NOTEARS: Controlling the graph prediction via variance manipulation', arXiv preprint arXiv:2206.07195.
- [3] Gao, M., Ding, Y. and Aragam, B. (2020), 'A polynomial-time algorithm for learning nonparametric causal graphs', Advances in Neural Information Processing Systems 33, 11599–11611.